

Los índices abajo $\rightarrow$ Jíndice
 arriba $\rightarrow$ Columna

Sección 3.7.4 Ejercicio 6

a
$$e^i = \frac{e_j \cdot x e_k}{e_i \cdot (e_j \cdot x e_k)}$$

$i, j, k = 1, 2, 3$ } sus permutaciones
 ciclo

imporemos que: $e^i \cdot e_j = \delta_{ij}$ Par sistema cíclico

$$e^i \cdot e_j = \frac{(e_j \cdot x e_k) \cdot e_i}{e_i \cdot (e_j \cdot x e_k)} = \frac{e_i \cdot (e_j \cdot x e_k)}{e_i \cdot (e_j \cdot x e_k)} = 1$$

$$e^i \cdot e_j = \frac{e_j \cdot (e_i \cdot x e_k)}{e_i \cdot (e_j \cdot x e_k)} = 0 \quad \text{ya que} \quad \begin{aligned} a &= e_j \\ a^\perp &= e_i \cdot x e_k \\ a \cdot a^\perp &= 0 \end{aligned}$$

Norma

Continuación 6

$$b. \quad V = e_1 \cdot (e_2 \times e_3) \quad \tilde{V} = e^1 \cdot (e^2 \times e^3)$$

$$V \tilde{V} = (e_1 \cdot (e_2 \times e_3)) (e^1 \cdot (e^2 \times e^3))$$

$$= (e_1^i \epsilon_{ijk} e_2^j e_3^k) (e^1_l \epsilon^{lmn} e_2^l e_3^m)$$

$$= e_1^i e_2^j e_3^k \epsilon_{ijk} \epsilon^{lmn} e_2^l e_3^m e_1^1$$

$$= (\delta_j^1 \delta_k^m - \delta_k^1 \delta_j^m) e_2^j e_3^k e_1^1 e_m^1$$

$$= \delta_j^1 \delta_k^m e_2^j e_3^k e_1^1 e_m^1 - \delta_k^1 \delta_j^m e_2^j e_3^k e_1^1 e_m^1$$

$$= e_2^j e_3^k e_1^1 e_k^1 - e_2^j e_3^k e_1^1 e_j^1 = 1 + 0 = 1$$

$$V \tilde{V} = 1 //$$

$$c. \quad a \cdot e^i = 1, \quad i=1,2,3$$

$$a = d^j e_j \Rightarrow d^j (e_j \cdot e^i) = d^j \delta_j^i = d^i$$

$$\text{Para } \forall i \quad a \cdot e^i = 1, \quad d^i = 1 \Rightarrow a = e_j //$$

Unicidad Supongamos que existen dos vectores a y b que cumplen: $a \cdot e^i = 1$, $b \cdot e^i = 1$

$$\text{Definimos } V = a - b = V^i e_i$$

$$V \cdot e^i = (a - b) \cdot e^i = a \cdot e^i - b \cdot e^i = 1 - 1 = 0$$

$$V \cdot e^i = V^j (e_j \cdot e^i) = V^j \delta_j^i = V^i$$

$$\text{Por lo tanto } V^i = 0 \Rightarrow a - b = 0$$

$$a = b$$

Conclusión: El vector a es único.

Continuación 6

1.1 $W_1 = 4\hat{i} + 2\hat{j} + \hat{k}$ $W_2 = 3\hat{i} + 3\hat{j}$ $W_3 = 2\hat{k}$

Base recíproca $\{e^i\}$

$e^i = \frac{e_j \times e_k}{e_i \cdot (e_j \times e_k)}$ Usamos la expresión denotada anteriormente

$W^1 = \frac{W_2 \times W_3}{W_1 \cdot (W_2 \times W_3)}$ $W_2 \times W_3 = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$

$W_1 \cdot (W_2 \times W_3) = \begin{pmatrix} 6 \\ 6 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 6\hat{i} - 6\hat{j} = \begin{pmatrix} 6 \\ -6 \\ 0 \end{pmatrix}$

① $W^1 = (1/2, -1/2, 0)$

$W^2 = \frac{W_3 \times W_1}{W_2 \cdot (W_3 \times W_1)}$ $W_3 \times W_1 = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ 4 & 2 & 1 \end{vmatrix} = \begin{pmatrix} -4 \\ 8 \\ 0 \end{pmatrix}$

$W_2 \cdot (W_3 \times W_1) = -12 + 24 = 12$

$W^2 = (-1/3, 2/3, 0)$ ②

$W^3 = \frac{W_1 \times W_2}{W_3 \cdot (W_1 \times W_2)}$ $W_1 \times W_2 = \begin{vmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 3 & 3 & 0 \end{vmatrix} = \begin{pmatrix} 3 \\ 3 \\ 6 \end{pmatrix}$

$W_3 \cdot (W_1 \times W_2) = 12$

$W^3 = (1/4, 1/4, 1/2)$ ③

Base recíproca $\{e^i\}$

$W^1 = (1/2, -1/2, 0)$ $W^2 = (-1/3, 2/3, 0)$

$W^3 = (1/4, 1/4, 1/2)$

$$11. 7 + 2\gamma + 3\beta = \alpha(4\gamma + 2\gamma + 1) + \beta(3\gamma + 3\gamma) + \phi(2\gamma)$$

$$7 = 4\alpha + 3\beta$$

$$2 = 2\alpha + 3\beta$$

$$3 = \alpha + 2\phi$$

$$3 + \frac{1}{2} = 2\phi \rightarrow \phi = \frac{7}{4}$$

Contraintes

$$\alpha = -\frac{1}{2}$$

$$\beta = 1 \quad \phi = \frac{7}{4}$$

$$a_i = a \cdot e^i$$

$$a_1 = (1, 2, 3) \cdot (-1/2, -1/2, 0)$$

$$a_1 = \frac{1}{2} - 1 + 0 = -\frac{1}{2}$$

$$a_2 = (1, 2, 3) \cdot (-1/3, 2/3, 0)$$

$$a_2 = -\frac{1}{3} + \frac{4}{3} + 0 = 1$$

$$a_3 = (1, 2, 3) \cdot (-1/4, 1/4, 1/2)$$

$$a_3 = -\frac{1}{4} + \frac{1}{2} + \frac{3}{2} = \frac{7}{4}$$

$$\text{Contraintes} \rightarrow \alpha^* = -\frac{1}{2} \quad \beta^* = 1 \quad \phi^* = \frac{7}{4}$$

San la
memo

① Es fácil ver que $\langle \vec{v}_i | \vec{v}_i \rangle = 2\delta_i$
ahora imponemos lo siguiente $\langle \vec{v}_i | \vec{v}'_i \rangle = \delta_i$

$$\langle \vec{v}_i | \vec{v}_i \rangle = 2\delta_i \rightarrow \frac{1}{2} \langle \vec{v}_i | \vec{v}_i \rangle = \delta_i$$

$$T \langle \vec{v}_i | \vec{v}_i \rangle = \langle \vec{v}'_i | \vec{v}_i \rangle$$

$$\rightarrow \langle \frac{1}{2} \vec{v}_i | \vec{v}_i \rangle = \langle \vec{v}'_i | \vec{v}_i \rangle$$

$$\text{Por lo tanto } \frac{1}{2} \vec{v}_i = \vec{v}'_i$$

Dado un vector general $|H\rangle = a_i \vec{v}_i$

Sabemos que la componente correspondiente está dada por

$$C_i = \langle H | \vec{v}_i \rangle$$

$$\text{Entonces: } \langle H | = C_i \vec{v}'_i = \langle H | \vec{v}_i \rangle \cdot \frac{1}{2} \vec{v}_i$$

Siendo esta su forma asociada